

Instruction Manual

HI 93828 Nitrate ISM



Dear Customer,

Thank you for choosing a Hanna product. This manual will provide you with the necessary information for the correct operation of the meter. Please read it carefully before using the meter. If you need additional technical information, do not hesitate to e-mail us at tech@hannainst.com.

This instrument is in compliance with CE directives.

Preliminary Examination

Remove the instrument from the packing material and examine it carefully to make sure that no damage has occurred during shipment. If there is any damage, notify your Dealer.

Each Ion Specific Meter is supplied complete with

- 9V Battery
- Two Sample Cuvets and Caps
- One Transport Cap

Note: Conserve all packing material until the instrument has been observed to function correctly. Any defective item must be returned in its original packing.

General Description

The HI 93828 meter measures the nitrate (NO_3^-) content in water and wastewater in the 0 to 100 mg/L (ppm) range.

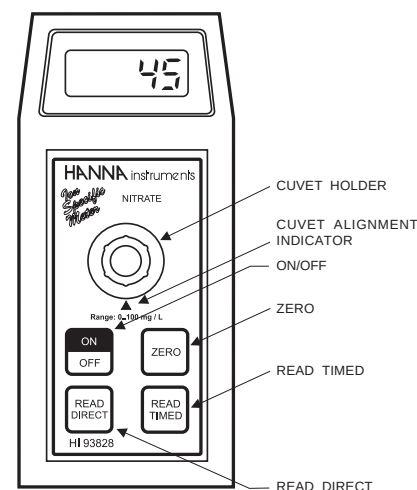
The meter uses an exclusive positive-locking system to ensure that the cuvet is in the same position every time it is placed into the measurement cell.

The reagent is in powder form and is supplied in packets. The amount of reagent is precisely dosed to ensure maximum repeatability.

Display codes aid the user in routine operations.

The meters have an auto-shut off feature that will turn the instrument off after 10 minutes of non-use.

Specifications



SPECIFICATIONS

Range	0 to 100 mg/L
Resolution	1 mg/L
Accuracy	$\pm 5 \text{ mg/L} \pm 5\%$ of reading
Typical EMC Deviation	$\pm 1 \text{ mg/L}$
Light Source	Light Emitting Diode @ 555 nm
Method	Adaptation of the cadmium reduction method. The reaction between nitrate and the reagent causes an amber tint in the sample
Light Detector	Silicon Photocell
Environment	0 to 50°C (32 to 122°F); max 95% RH non-condensing
Battery Type/Life	1 x 9 volt/40 hours
Auto-Shut off	After 10' of non-use
Dimensions	180 x 83 x 46 mm (7.1 x 3.3 x 1.8")
Weight	290 g (10 oz.).

REQUIRED REAGENTS

Code	Description	Quantity
HI 93728-0	Nitrate Reagent	1 packet

REAGENT SETS

HI 93728-01	Reagents for 100 tests
HI 93728-03	Reagents for 300 tests

Display Code Guide

This indicates that the meter is in a ready state and zeroing can be performed.

5 IP

Sampling in Progress. This prompt appears each time the meter is performing a measurement.

-00-

This indicates that the meter is in a zeroed state and measurement can be performed.

2ER0

A zero reading was not taken. Insert a sample before adding reagent and press ZERO.

0

Under range. A blinking "0" indicates that the sample absorbs less light than the zero reference. Check the procedure and make sure you use the same cuvet for reference (zero) and measurement.

100

Over range. A flashing value higher than the maximum concentration readable (see specifications) indicates that the sample absorbs too much light, meaning that the concentration is too high. Dilute the sample.

CAP

Light over range. The cuvet is not inserted correctly and an excess ambient light is reaching the detector. If the cover is properly installed, then contact your dealer or the nearest Hanna Customer Service Center.

L0

Light under range. The zero sample is too dark for proper zeroing. If this is not the case, contact your dealer or the nearest Hanna Customer Service Center.

V 10

The "V" indicates that the battery voltage is getting low and the battery needs to be replaced.

-BA-

This indicates that the battery is dead and must be replaced.

Note: once this indication is displayed, the meter will lockup. Change the battery to restart.



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CE
This Instrument is in
Compliance with the CE Directives

Warranty

HI 93828 is warranted for two years against defects in workmanship and materials when used for its intended purpose and maintained according to instructions.

This warranty is limited to repair or replacement free of charge.

Damages due to accident, misuse, tampering or lack of prescribed maintenance are not covered.

If service is required, contact the dealer from whom you purchased the instrument. If under warranty, report the model number, date of purchase, serial number and the nature of the failure. If the repair is not covered by the warranty, you will be notified of the charges incurred. If the instrument is to be returned to Hanna Instruments, first obtain a Returned Goods Authorization Number from the Customer Service department and then send it with shipment costs prepaid. When shipping any instrument, make sure it is properly packaged for complete protection.

To validate your warranty, fill out and return the enclosed warranty card within 14 days from the date of purchase.

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HANNA instruments® reserves the right to modify the design, construction and appearance of its products without advance notice.

Operational Guide

MEASUREMENT PROCEDURE

- Turn the meter on by pressing ON/OFF.



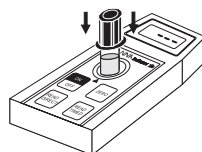
- When the LCD displays "---", it is ready.



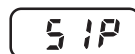
- Add to the cuvet 6 mL of sample, up to about half of its height, and replace the cap.



- Place the cuvet into the holder and ensure that the notch on the cap is positioned securely into the groove.



- Press ZERO and "SIP" will appear on the display.



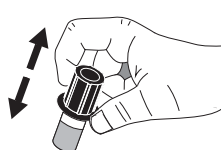
- Wait for a few seconds and the display will show "-0.0-". Now the meter is zeroed and ready for measurement.



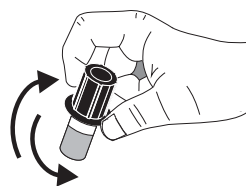
- Remove the cuvet and add the content of one packet of HI 93728-0 Nitrate Reagent.



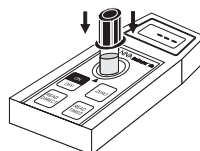
- Replace the cap and immediately shake vigorously for exactly 10 seconds by moving the cuvet up and down.



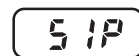
Continue to mix by inverting the cuvet gently and slowly for 50 seconds, while taking care not to induce air bubbles. A deposit remains but it does not affect the measurement. Time and way of shaking could sensitively affect the measurement.



- Reinsert the cuvet into the instrument, taking care not to shake it.



- Press READ TIMED and the display will show the countdown prior to the measurement or, alternatively, wait for 4 minutes and 30 seconds and press READ DIRECT. In both cases "SIP" will appear during measurement.



- The instrument directly displays concentration in mg/L of nitrate on the Liquid Crystal Display.
- To convert the reading to mg/L of nitrate-nitrogen ($\text{NO}_3\text{-N}$), multiply by a factor of 0.226.

INTERFERENCES

Interference may be caused by:

Ammonia and amines, as urea and primary aliphatic amines
Chloride above 100 ppm (negative interference)
Chlorine above 2 ppm (positive interference)
Copper (it must be absent)
Iron (III) (positive interference)
Strong oxidizing and reducing substances
Sulfide (it must be absent)

Note: To ensure accurate results, perform the tests at room temperature, between 18°C and 28°C (65°F to 83°F).

For best results: Intensely colored samples or suspended matter in large amounts will cause interference. Therefore sample should be adequately treated before performing the test.

Tips for an Accurate Measurement

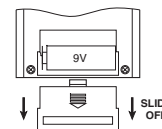
The instruction listed below should be carefully followed during testing to ensure best accuracy.

- Do not touch the cuvet walls with hands.
- In order to maintain the same conditions during the zeroing and the measuring phases, it is necessary to close the cuvet to prevent any contamination.
- Do not let the test sample stand too long after reagent is added or accuracy will be lost.
- Whenever the cuvet is placed into the measurement cell, it must be completely free of fingerprints, oil or dirt. Wipe it thoroughly with HI 731318 or a lint-free cloth prior to insertion.
- It is important that the sample does not contain any debris. This would corrupt the readings.
- It is possible to take multiple readings in a row, but it is recommended that a zero reading be taken for each sample and that the same cuvet is used for zeroing and measurement.
- It is important to discard the sample immediately after the reading is taken because the glass may become permanently stained.
- Shaking the cuvet can generate bubbles in the sample, causing higher readings. To obtain accurate measurements, remove such bubbles by swirling or by gently tapping the vial.
- All the reaction times reported in this manual are referred to 20°C (68°F). As a general rule, they should be doubled at 10°C (50°F) and halved at 30°C (86°F).

Battery Replacement

Battery replacement must only take place in a non-hazardous area using a 9V alkaline battery.

Simply slide off the battery cover on the back of the meter. Detach the battery from the terminals and attach a fresh 9V battery while paying attention to the correct polarity. Replace the battery and the cover.



Accessories

REAGENT SETS

- HI 93728-01 Reagents for 100 tests
- HI 93728-03 Reagents for 300 tests

OTHER ACCESSORIES

- HI 710009 Blue rubber boot
- HI 710010 Orange rubber boot
- HI 721310 9V battery (10 pcs)
- HI 731318 Tissue for wiping cuvetts (4 pcs)
- HI 731321 Glass cuvetts (4 pcs)
- HI 731325 Caps for cuvetts (4 pcs)
- HI 93703-50 Cuvets cleaning solution (230 mL)

CE Declaration of Conformity



DECLARATION OF CONFORMITY

We

Hanna Instruments Italia Srl
viale Delle Industrie, 12/A
35010 Villafranca Padovana - PD
ITALY

herewith certify that the meter:

HI 93828

Has been tested and found to be in compliance with EMC Directive 89/336/EEC and Low Voltage Directive 73/23/EEC according to the following applicable normative:

EN 50082-1: Electromagnetic Compatibility - Generic Immunity Standard
IEC 61000-4-2 Electrostatic Discharge
IEC 61000-4-3 RF Radiated

EN 50081-1: Electromagnetic Compatibility - Generic Emission Standard
EN 55022 Radiated, Class B

EN 61010-1: Safety requirements for electrical equipment for measurement, control and laboratory use

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D. Volpato - Engineering Manager
On behalf of
Hanna Instruments Italia S.r.l.

Recommendations for Users

Before using these products, make sure that they are entirely suitable for the environment in which they are used.

Operation of these instruments in residential area could cause unacceptable interferences to radio and TV equipments, requiring the operator to take all necessary steps to correct interferences.

Any variation introduced by the user to the supplied equipment may degrade the instruments' EMC performance.

To avoid damages or burns, do not perform any measurement in microwave ovens.

Safety Data Sheets

Read the relevant SDS sheets before performing the test.

Safety Data Sheets are available at: www.hannainst.com